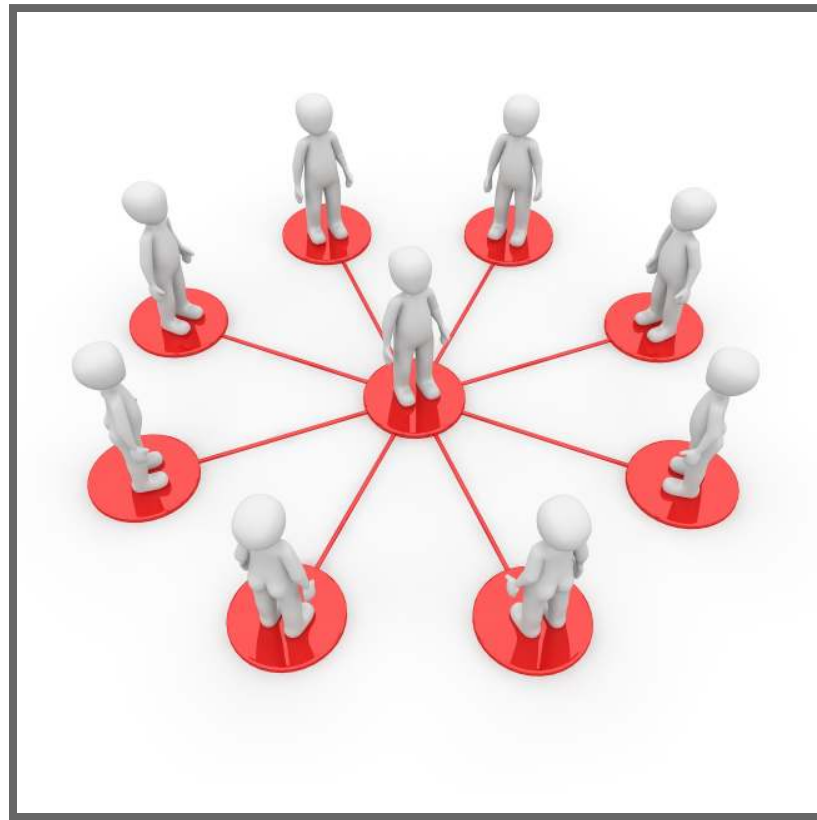




Building Multiplayer Games

with Phoenix and Phaser

Created by [Keith Salisbury](#) / [@ktec](#)



Fun Stack

The fun way to build scalable social games



Designed for
web, desktop,
mobile with
amazing tutorials
and great
documentation

Phoenix Framework

Productive. Reliable. Fast

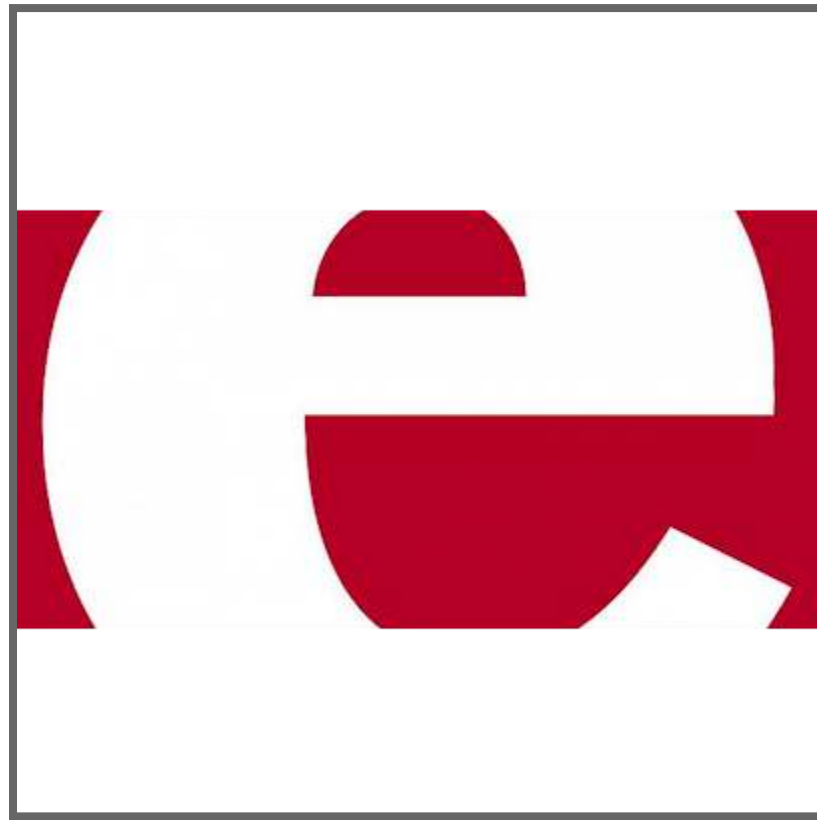
A productive web framework that does not compromise **speed** and maintainability



Elixir

dynamic, functional, meta-programming aware

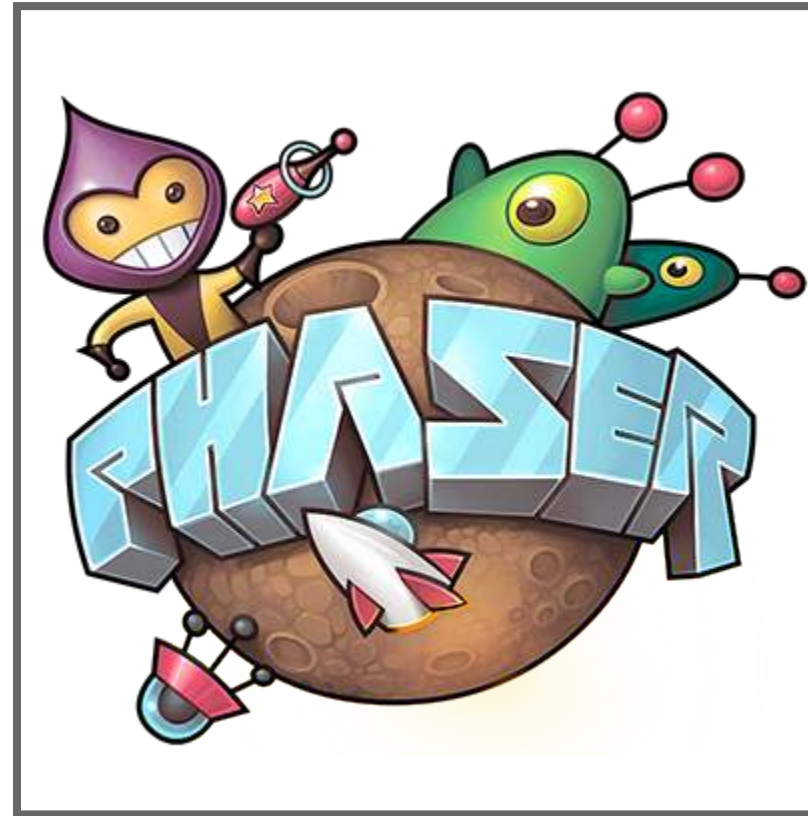
designed for building scalable and maintainable applications



Erlang

low-latency, distributed and fault-tolerant

build massively scalable soft real-time systems with requirements on high availability



Designed for
web, desktop,
mobile with
amazing tutorials
and great
documentation

Phaser Game Framework

The fun, fast and free HTML5 Game Framework



Pixi Rendering Engine

Super **fast** HTML 5 2D rendering engine that uses WebGL with canvas fallback



HTML5

Offline Storage, Device Access, **Web Sockets**
Multimedia, Graphics and Effects, Web Workers, CSS3

Features

Phoenix

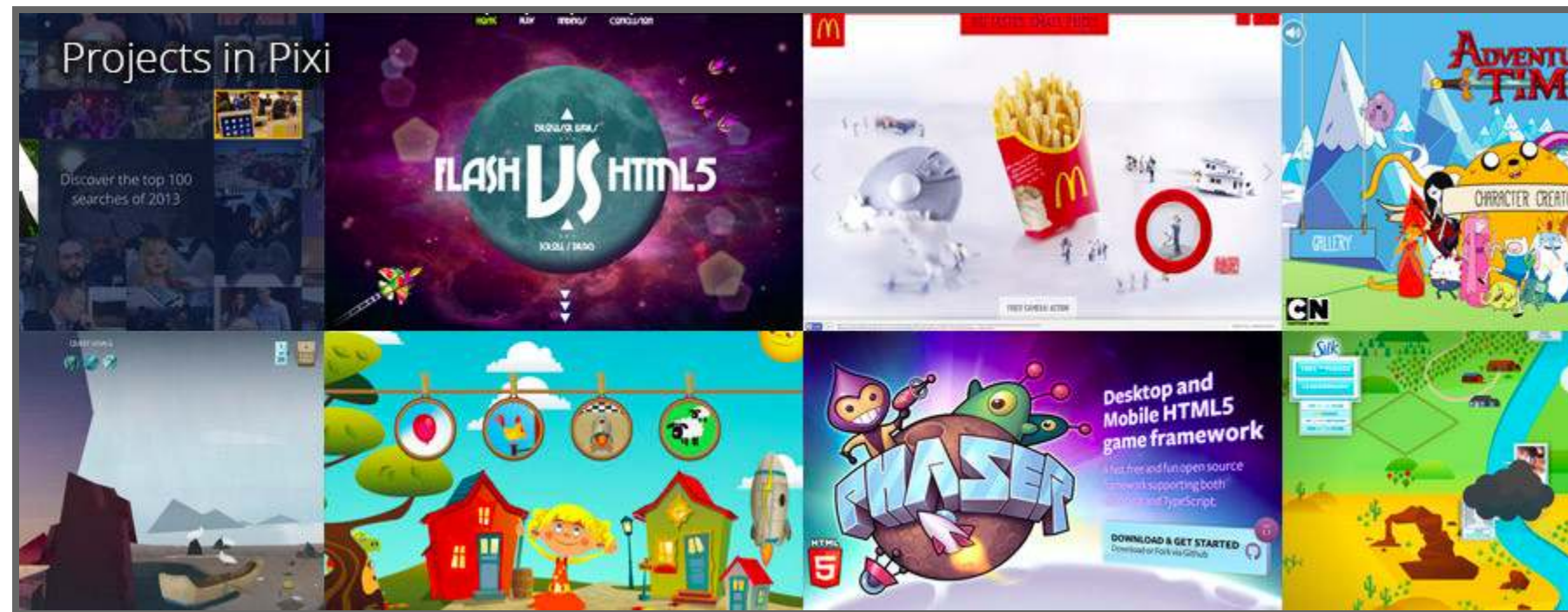
easy - scalable - fast - friendly - channels - ES6/7 ready

Elixir

syntax - metaprogramming - macros - tooling

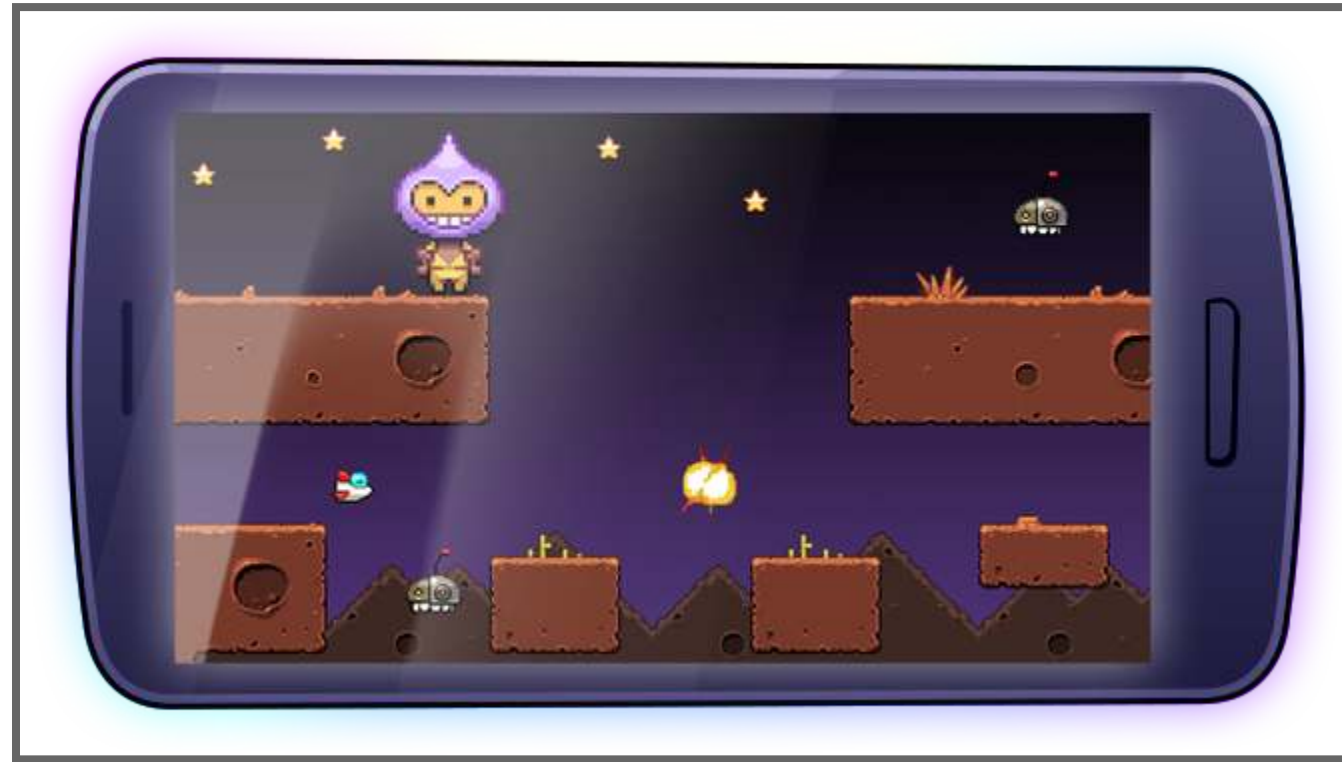
Erlang/OTP

semantics - fault tolerant - scalable - distributed



Pixi

ultra fast - multi platform - multi touch - sprite sheet
asset loader - auto-detect (webgl or canvas) - rich text



Phaser

preloader - [physics](#) - groups - [sprites](#) - camera

particles - input - [sound](#) - tile maps - scale manager

Pointless Social Stats



This repository Search

Pull requests Issues Gist



html5 / spec

👁 Watch

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23

k 2

↔ Code

🔔 Issues 1

🔗 Pull requests 0

📖 Wiki

📈 Pulse

📊 Graphs

The HTML5 Specification - specification editing and generation tools <http://whatwg.org/>

📦 38 commits

🌿 2 branches

📦 0 releases

👤 2 contributors

Branch: master ▾

New pull request

New file

Upload files

Find file

SSH ▾

git@github.com:html5/spec



Download ZIP



msporny Added perma-location for todays Editors Draft for HTML5 Profiles.

Latest commit 98bdd92 on Apr 17, 2010



bin

Fixed minor bugs when inserting bugs against status-of-this-document.

6 years ago



configs

Integrating changes from Larry Masinter into latest HTML+RDFa WD.

6 years ago



docs

Updated and released 2nd HTML+RDFa draft.

7 years ago



drafts/ED-html5-profiles-20040417

Added perma-location for todays Editors Draft for HTML5 Profiles.

6 years ago



headers

Fixed minor bugs when inserting bugs against status-of-this-document.

6 years ago



microsections

Added Infoset and DOM2 specifics to HTML5+RDFa draft.

6 years ago



setup

Added basic configure.ac script, Makefile.in and support scripts for

7 years ago



source

Added HTML5-profiles specification.

6 years ago



utils

Added Anolis and anolislib to the build utilities.

7 years ago



Makefile.in

Added HTML5-profiles specification.

6 years ago



README

Fixed small documentation error in README.

7 years ago

Anyway...

Lets see some code

I HATE

JAVASCRIPT



Hello World

- Create Phoenix project
- Install Phaser
- Create a game
- Update the HTML page
- Start the server
- Create a game state
- Add text "Hello world"
- Make it draggable

Create a new Phoenix project...

```
$ mix phoenix.new demo
```

```
...
```

```
$ cd demo
```


Install Phaser...

```
$ mkdir web/static/vendor/js/phaser  
$ curl -L -o web/static/vendor/js/phaser/phaser  
https://raw.githubusercontent.com/photonstorm/p
```

* Use vendor so Brunch doesn't pre-compile

Create a Game class...

```
export class Game extends Phaser.Game {  
    // Initialize Phaser  
    constructor(width, height, container) {  
        super(width, height, Phaser.AUTO, container  
    }  
}
```

web/static/js/Game.js

Create new Game instance...

```
// Import dependencies
import {Game} from "../Game"

// Lets go!
new Game(700, 450, "phaser",
```



web/static/js/app.js

Set up the page template...

```
<div id="phaser">v>
```

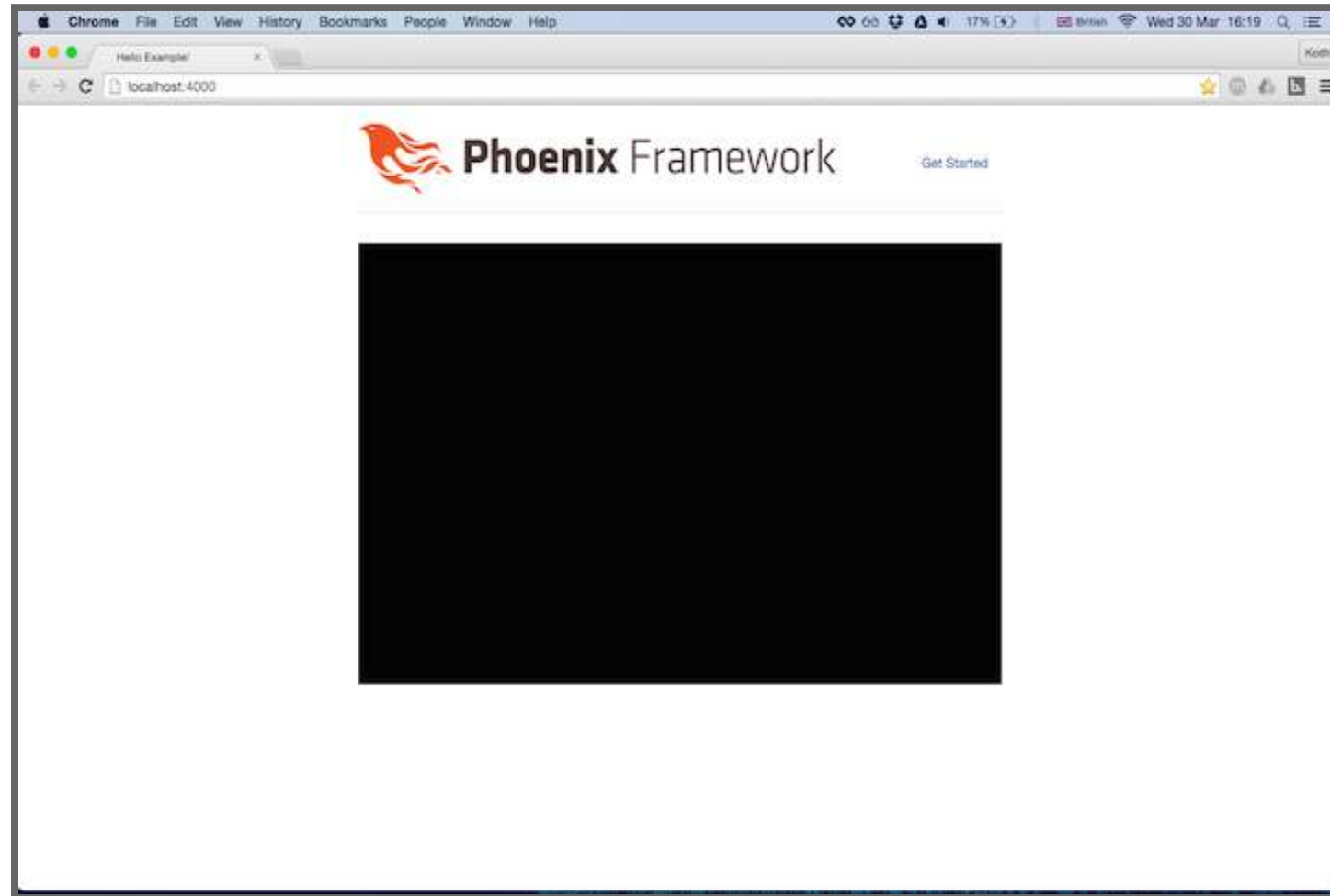


web/templates/page/index.html.eex

Start up the server...

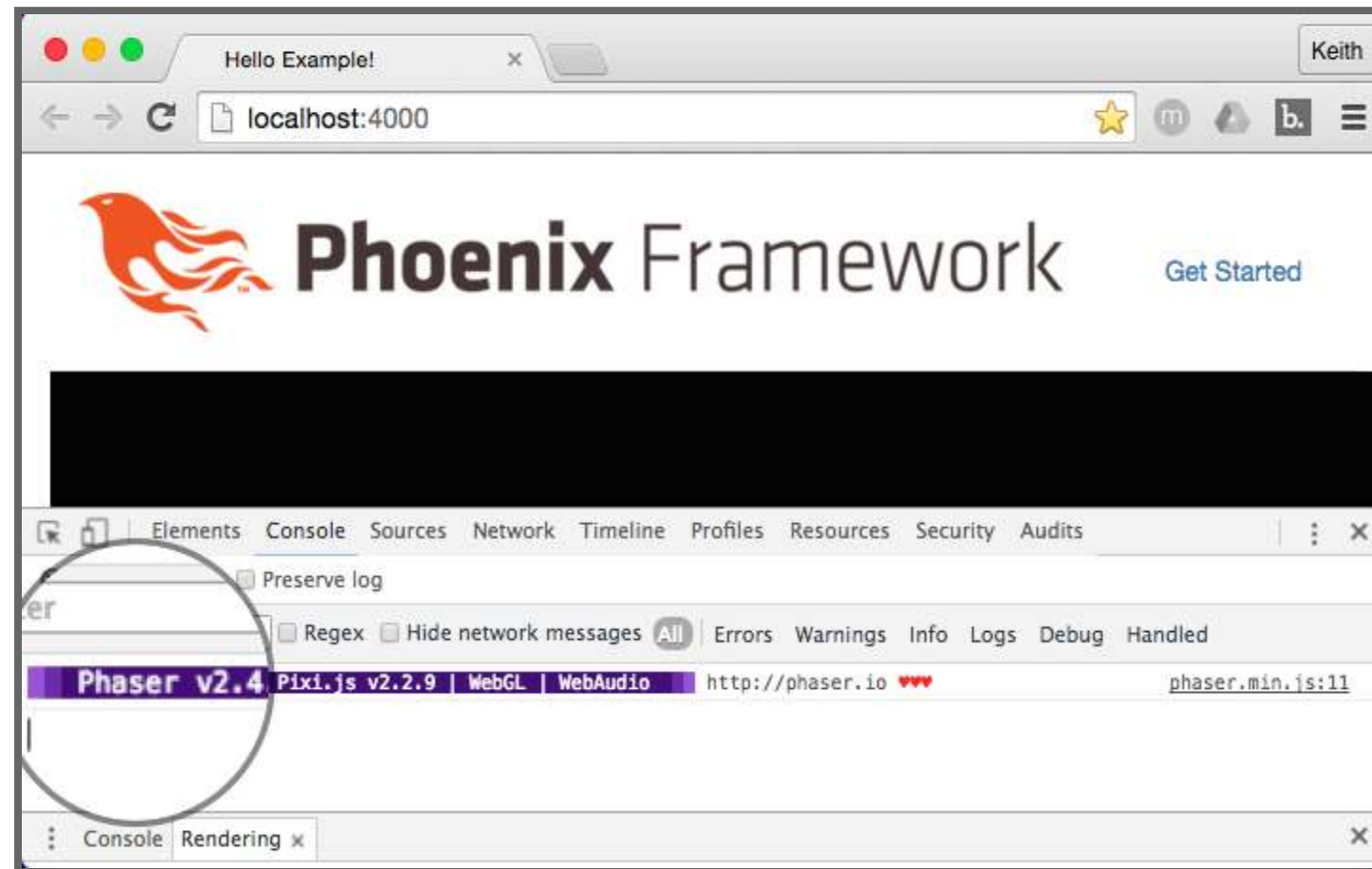
```
$ iex -S mix phoenix.server
```

Boom!



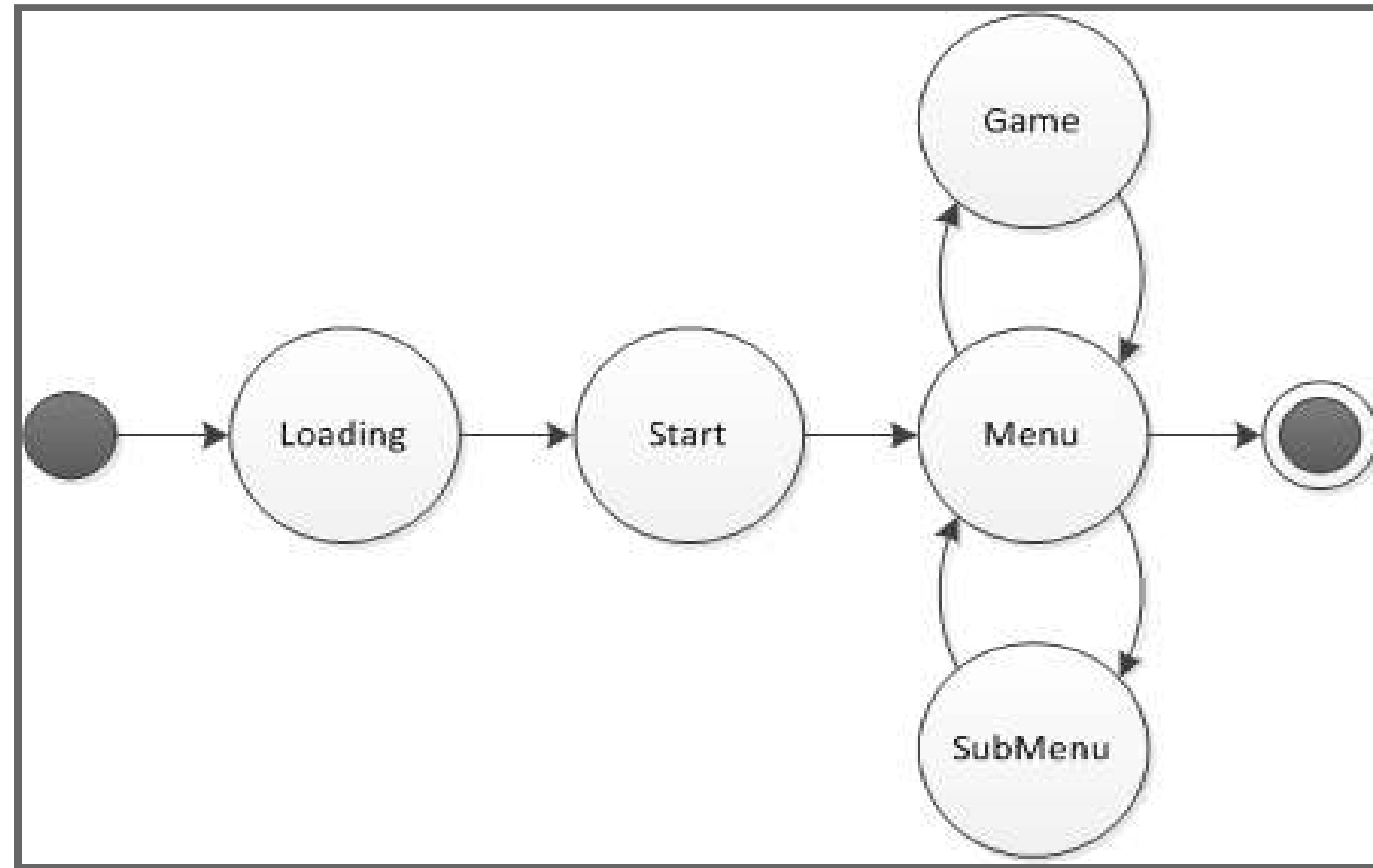
But wait?

Open the console



Where's my "Hello world"?

Game States



Using states allows you to break your game up into smaller pieces that can handle different mechanics of the game, such as a menu.

Create states directory

```
$ mkdir web/static/js/states
```


Create a new "State"

```
import { createLabel } from "../common/labels"

export class Lobby extends Phaser.State {
  create() {
    const label = createLabel(this, "Hello world")
    label.anchor.setTo(0.5)
  }
}
```

web/static/js/states/**Lobby.js**

Nothing up my sleeves

```
const DEFAULT_STYLE = {font: "65px Arial", fill: "#ffffff" }

// createLabel :: State -> String -> Object -> Sprite
export const createLabel = (state, message, style = DEFAULT_STYLE)

  const {centerX, centerY} = state.world
  return state.add.text(centerX, centerY, message, style)

}
```

web/static/js/common/[labels.js](#)

Update the Game class

```
import {Lobby} from "../states/Lobby"

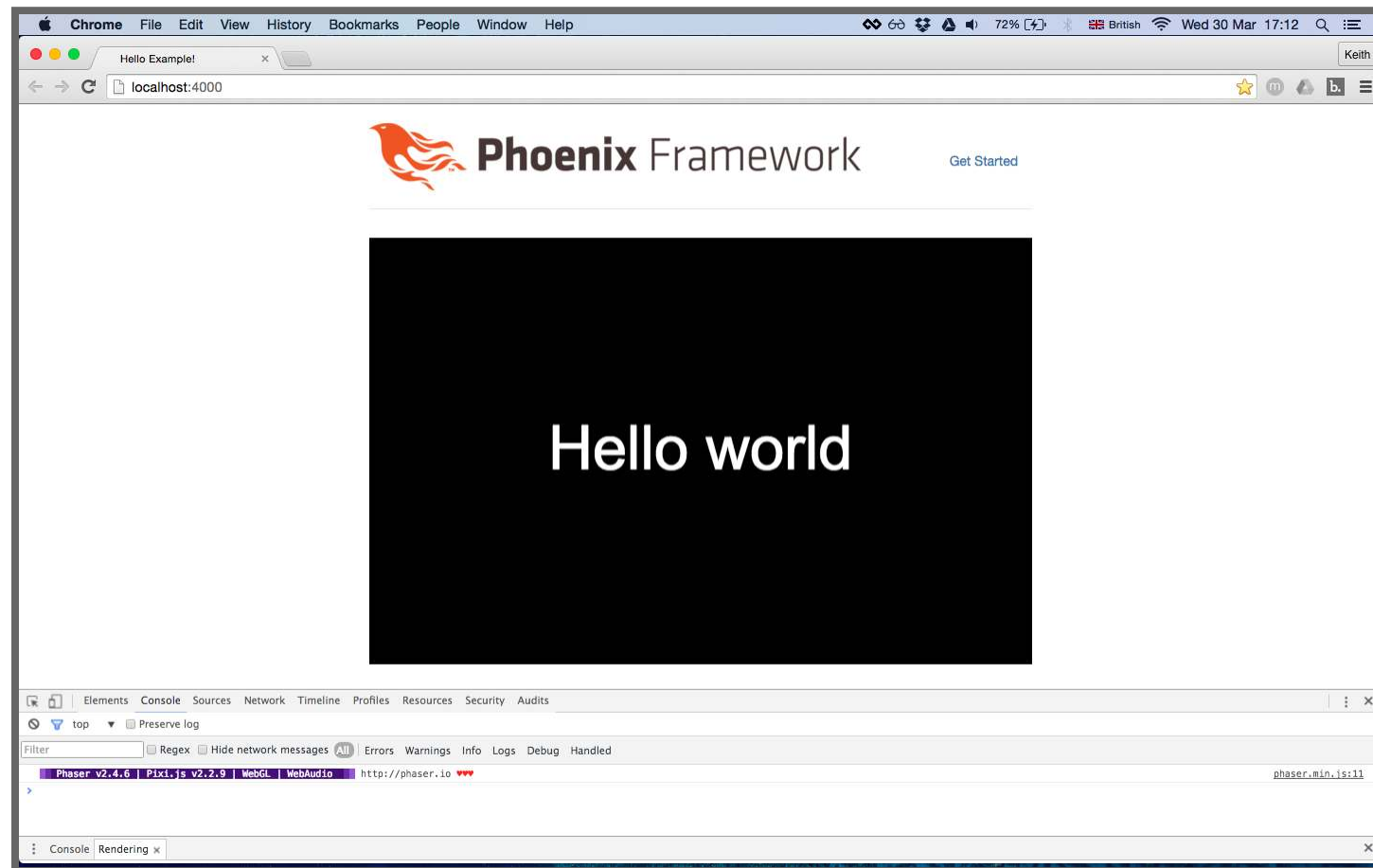
export class Game extends Phaser.Game {
  constructor(width, height, container) {
    super(width, height, Phaser.AUTO, container)

    this.state.add("lobby", Lobby, false)

    this.state.start("lobby")
  }
}
```

web/static/js/Game.js

Yay!



But wait?

We can do better

Lets make the text draggable

```
export class Lobby extends Phaser.State {  
  create() {  
    const label = createLabel(this, "Hello world")  
    label.anchor.setTo(0.5)  
    label.inputEnabled = true  
    label.input.enableDrag()  
  }  
}
```

web/static/js/states/Lobby.js



Phoenix Framework

[Get Started](#)

Hello world

Whats next?

Synchronised text?

Because `\_(ツ)\_/`



Phoenix Channels

Multiplexed on HTML5 WebSockets

Built in

Asynchronous

Create a new Phoenix channel

```
$ mix phoenix.gen.channel Lobby games  
* creating web/channels/lobby_channel.ex  
* creating test/channels/lobby_channel_test.exs
```

Add the channel to your `web/channels/user_socket.ex` handler, for

```
channel "games:lobby", Demo.LobbyChannel
```

Phoenix creates this...

```
defmodule Demo.LobbyChannel do
  use Demo.Web, :channel

  def join("games:lobby", payload, socket) do
    if authorized?(payload) do
      {:ok, socket}
    else
      {:error, %{reason: "unauthorized"}}
    end
  end

  def handle_in("shout", payload, socket) do
    broadcast socket, "shout", payload
    {:noreply, socket}
  end

  ...
end
```

web/channels/lobby_channel.ex

Add the channel to the UserSocket

```
defmodule Demo.UserSocket do
  use Phoenix.Socket

  ## Channels
  channel "games:lobby", Demo.LobbyChannel
  ...
end
```

web/channels/user_socket.ex

Pass the socket to the game

```
// Import dependencies
import {Game} from "./Game"
import {Socket} from "phoenix"

const socket = new Socket("/socket", {})
const game = new Game(700, 450, "phaser")

// Lets go!
game.start(socket)
```

web/static/app.js

Connect the socket

```
export class Game extends Phaser.Game {  
  constructor(width, height, container) {  
    super(width, height, Phaser.AUTO, container, null)  
  
    // set up game states  
    this.state.add("lobby", Lobby, false)  
  }  
  
  start(socket) {  
    socket.connect()  
  }  
}
```

web/static/js/Game.js

Create the channel

```
import {joinChannel} from "../common/channels"

export class Game extends Phaser.Game {
  ...
  start(socket) {
    socket.connect()

    // create and join the lobby channel
    const channel = socket.channel("games:lobby", {})

    joinChannel(channel, () => {
      console.log("Joined successfully")
      // start the lobby [name, clearWorld, clearCache, ...stateIni
      this.state.start("lobby", true, false, channel)
    })
  }
}
```

web/static/js/Game.js

Log the channel reference

```
export class Lobby extends Phaser.State {  
  init(...args) {  
    const [channel] = args  
    console.log(channel)  
  }  
}
```

web/static/js/Lobby.js

Inside channels.js

```
// joinChannel :: Channel -> Channel
export const joinChannel = (channel, success, failure, timeout) =>
  channel
    .join()
    .receive("ok", success || joinOk)
    .receive("error", failure || joinError)
    .receive("timeout", timeout || joinTimeout)
  return channel
}

// joinOk :: Response -> Console
const joinOk = (response) => console.log(`Joined successfully`, res

// joinError :: Response -> Console
const joinError = (response) => console.log(`Failed to join channel

// joinError :: Null -> Console
const joinTimeout = () => console.log("Networking issue. Still wait
```




Phoenix Framework

[Get Started](#)

Hello world

Phaser v2.4.7 | Pixi.js v2.2.9 | WebGL | WebAudio http://phaser.io ❤❤❤ phaser.js:35027
Joined successfully channels.js:7
▶ Channel {state: "joined", topic: "games:lobby", params: Object, socket: Socket, bindings: Array[3]...} Lobby.js:6

Ok nearly there

lets send the position when we stop dragging

Store the channel reference

```
export class Lobby extends Phaser.State {  
  init(...options) {  
    const [channel] = options  
    this.channel = channel  
  }  
}
```

web/static/js/Lobby.js

Make the label sync

```
import { createSyncLabel } from "../common/sync_labels"

export class Lobby extends Phaser.State {
  ...
  create() {
    const label = createSyncLabel(this,
                                  "Hello world",
                                  this.channel)
  }
}
```

web/static/js/Lobby.js

And here's the sync label

```
import { createLabel } from "../labels"
import { syncPosition } from "../sync"

export const createSyncLabel = (state, message, channel) => {
  const label = createLabel(state, message)
  label.anchor.setTo(0.5)
  label.inputEnabled = true
  label.input.enableDrag()

  // send message on drag stop [sprite, channel, event]
  syncPosition(label, channel, label.events.onDragStop)

  return label
}
```

web/static/js/common/sync_labels.js

And finally the sync position

```
// syncPosition :: Sprite -> Channel -> Event -> Function -> Event
export const syncPosition = (sprite, channel, event) => {
  event.add(sprite => sendPosition(sprite, channel))
}

// sendPosition :: Sprite -> Channel -> String
export const sendPosition = (sprite, channel) => {
  console.log(serializePosition(sprite))
}

// serializePosition :: Sprite -> Object
export const serializePosition = ({x, y}) => Object.assign({x, y})
```

<web/static/js/common/sync.js>

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Hello Demo! x Keith

localhost:4000



Phoenix Framework

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Hello world

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Joined successfully channels.js:7

Console

100% British Fri 6 May 13:51

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Joined successfully channels.js:7

Console

We're ready to
SEND the message

Send the message

```
// sendPosition :: Sprite -> Channel -> Push
export const sendPosition = (sprite, channel) => {
  const message = serializePosition(sprite)
  console.log("Sending message", message)
  channel.push("shout", message)
}
```

<web/static/js/common/sync.js>

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Hello Demo! x Keith

localhost:4000



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localhost:4000



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Joined successfully channels.js:7

Console

We're ready to
RECEIVE the message

Receive the message

```
// syncPosition :: Sprite -> Channel -> Event -> Function -> Event
export const syncPosition = (sprite, channel, event) => {
  event.add(sprite => sendPosition(sprite, channel))
  receivePosition(sprite, channel)
}

// receivePosition = Sprite -> Channel -> Push
export const receivePosition = (sprite, channel) => {
  channel.on("shout", (message) => {
    console.log("Received message", message)
  })
}
```

web/static/js/common/sync.js

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Joined successfully channels.js:7

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Joined successfully channels.js:7

Console

Remember this?

```
defmodule Demo.LobbyChannel do
  ...
  def handle_in("shout", payload, socket) do
    broadcast socket, "shout", payload
    {:noreply, socket}
  end
  ...
end
```

[web/channels/lobby_channel.ex](#)

broadcast_from

Lets make a position handler

```
defmodule Demo.LobbyChannel do
  ...
  # broadcast position data to everyone else
  def handle_in("position", payload, socket) do
    broadcast_from socket, "position", payload
    {:noreply, socket}
  end
  ...
end
```

web/channels/lobby_channel.ex

Update the sync to use position

```
// sendPosition :: Sprite -> Channel -> Push
export const sendPosition = (sprite, channel) => {
  ...
  channel.push("position", message)
}

// receivePosition = Sprite -> Channel -> Push
export const receivePosition = (sprite, channel) => {
  ...
  channel.on("position", callback)
}
```

<web/static/js/common/sync.js>

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Hello Demo! x Keith

localhost:4000



Phoenix Framework

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Hello world

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Joined successfully channels.js:7

Console

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Hello Demo! x Keith

localhost:4000



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Joined successfully channels.js:7

Console

Remember to restart your server

```
[error] GenServer #PID<0.413.0> terminating
** (FunctionClauseError) no function clause matching in Demo.LobbyChannel
    (demo) web/channels/lobby_channel.ex:20:
    Demo.GameChannel.handle_in("position", %{ "x" => 249, "y" => 109
    %Phoenix.Socket{assigns: %{}, channel: Demo.LobbyChannel, channel
    #PID<0.413.0>, endpoint: Demo.Endpoint, handler: Demo.UserSocket
    joined: true, pubsub_server: Demo.PubSub, ref: "10", serializer
    Phoenix.Transports.WebSocketSerializer, topic: "games:lobby", t
    Phoenix.Transports.WebSocket, transport_name: :websocket, trans
    #PID<0.396.0>})
```

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Hello Demo! x Keith

localhost:4000



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Console

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Joined successfully channels.js:7

Console

Now finally we can update the sprite position!

```
// receivePosition = Sprite -> Channel -> Push
export const receivePosition = (sprite, channel) => {
  const callback = (message) => {
    console.log("Received message", message)
    const {x,y} = message
    sprite.position.setTo(x, y)
  }
  channel.on("position", callback)
}
```

<web/static/js/common/sync.js>

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localhost:4000



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Console

100% British Fri 6 May 16:09

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localhost:4000



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Phaser v2.4.7 | Pixi.js v2.2.9 | WebGL | WebAudio http://phaser.io ♥♥♥ phaser.js:35027
Joined successfully channels.js:7

Console

But wait?

We can do better

Use the onDragUpdate event

```
export const createSyncLabel = (state, message, channel) => {  
  ...  
  syncPosition(label, channel, label.events.onDragUpdate)  
  return label  
}
```

web/static/js/Lobby.js

Chrome File Edit View History Bookmarks People Window Help

Hello Demo! x Keith

localhost:4000



Phoenix Framework

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Console

100% British Fri 6 May 16:13

Hello Demo! x Keith

localhost:4000



Phoenix Framework

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Joined successfully channels.js:7

Console

Lets add some juice!



Phoenix Framework

[Get Started](#)

Phoenix Framework

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Clientside state problem



Phoenix Framework

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Hello world

Serverside State

Use a Genserver

But this is Erlang?

Lets use OTP

Supervision Trees



This repository Search

Pull requests Issues Gist



ktec / phoenixphaserdemo

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Building games with Phoenix and Phaser demo — Edit

19 commits 6 branches 0 releases 1 contributor

Branch: master New pull request New file Upload files Find file SSH git@github.com:ktec/phoen Download ZIP

Switch branches/tags

Find or create a branch...

Branches Tags

01_synchronised_text

02_uuid_user_token

03_multiple_channels

04_supervision_tree

05_rocks

✓ master

mix.exs

mix.lock

package.json

README.md

Latest commit fe87881 4 days ago

al commit with phoenix new (v1.1.4) 5 days ago

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gen channel 19 hours ago

nge to drag update event 12 hours ago

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initial commit with phoenix new (v1.1.4) 5 days ago

initial commit with phoenix new (v1.1.4) 5 days ago

initial commit with phoenix new (v1.1.4) 5 days ago

Identify each user using token

app.html.eex

```
5 <meta http-equiv="X-UA-Compatible"
  content="IE=edge">
6 <meta name="viewport" content="width=device-width,
  initial-scale=1">
7 <meta name="description" content="">
8 <meta name="author" content="">
9 <%= tag(:meta, name: "token", content: @token) %>
10
11 <title>Hello Demo!</title>
12 <link rel="stylesheet" href="<%=
  static_path(@conn, "/css/app.css") %>">
13 </head>
14
15 <body>
```

app.js

```
12 import {Socket} from 'phoenix'
13
14 const token =
15 document.head.querySelector("[name=token]").content
16 const socket = new Socket("/socket", {
17   params: {token: token},
18   // logger: (kind, msg, data) => {
19     console.log(`${kind}: ${msg}`, data)
20   }
21 })
22 const game = new Game(700, 450, "phaser")
23
24 // Lets go!
25 game.start(socket)
```



localhost:4000



Phoenix Framework

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Hello world



Elements

Console

Sources

Network

Timeline



```
<meta name="description" content>
<meta name="author" content>
<meta content=
  "g3QAAAACZAAEZGF0YW0AAAAkM2FkZjhlZWEtNTFjYy00YmZmLWI0M2EtNmY3ZmFmO
  TZmYmE3ZAAGc2lnbmVkbG9AYWDXUmlQB##48i6faB20WFFJ3ZJnsgzSr4nyco="
  name="token">
<title>Hello Demo!</title>
<link rel="stylesheet" href="/css/app.css">
</head>
```

... <body> == \$0

```
> <div class="container">...</div>
<!-- /container -->
```

html

body

Styles

Event Listeners

DOM Breakpoints

Properties

Multiple channels and Game states

play_channel.ex

```
1 defmodule Demo.PlayChannel do
2   use Demo.Web, :channel
3   require Logger
4
5   def join("games:play", payload, socket) do
6     if authorized?(payload) do
7       Logger.debug "#{socket.assigns.user_id} joined
8         the Play channel"
9       {:ok, socket}
10    else
11      {:error, %{reason: "unauthorized"}}
12    end
13  end
14 end
```

play_channel.ex

Play.js

```
1 import { createLabel } from "../common/labels"
2 import { createSyncLabel } from
3   "../common/sync_labels"
4 import { leaveChannel } from "../common/channels"
5
6 export class Play extends Phaser.State {
7   create(game) {
8     game.stage.backgroundColor = 0x551A8B
9
10    const label = createSyncLabel(this, "Move me!",
11      this.channel, "1")
12
13    const label2 = createLabel(this, "Click to return
```

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Move me!

Click to return to Lobby

Dynamic Processes and Supervision Trees

player_supervisor.ex

```
1 defmodule Demo.PlayerSupervisor do
2   alias Demo.Player
3
4   def start_link do
5     import Supervisor.Spec, warn: false
6     children = [
7       worker(Player, [], [restart: :transient])
8     ]
9     opts = [strategy: :simple_one_for_one,
10             max_restart: 0, name: __MODULE__]
11     Supervisor.start_link(children, opts)
12   end
13
14   def start(id) do
15     Supervisor.start_child(__MODULE__, [[id]])
16   end
17 end
```

player.ex

```
1 defmodule Demo.Player do
2   use GenServer
3   alias Demo.{Endpoint, Randomise}
4
5   defmodule State do
6     defstruct id: nil,
7               position: %{x: 0, y: 0},
8               type: "square"
9   end
10
11   ### PUBLIC API ###
```

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Move me!

Click to return to Lobby

Rock Paper Scissors



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Move me!
Click to Play



Phoenix Framework

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Move me!
Click to Play

github.com/ktec/phoenixphaserdemo

Message Flow



Even Elixir/Erlang/OTP is not *that*
fast

**Using broadcast_from! is not
enough**

So what to do?

One option is to have the server gather and send only one update,
and set a broadcast frequency for the node

```
def start_broadcasting do
  {:ok, _} = :timer.send_interval(200, :broadcast)
end

def handle_info(:broadcast, state) do
  # gather all the information the client needs here....
  broadcast_update(state)
  {:noreply, state}
end
```

Demos

rocks.globalkeith.com

dots.globalkeith.com

snake.globalkeith.com

Future

ElixirScript

```
defmodule Game do
  JS.import Menu, "./Elixir.Menu"

  def init() do
    game = JS.new Phaser.Game,
      [800, 600, Phaser.CANVAS, "phaser"]

    game.state.add("menu", %{
      "create" => &Menu.create/1,
      "update" => &Menu.update/1
    })

    game.state.start("menu")
  end
end
```

ElixirScript

```
def update(game) do
  Elixir.Enum.each.sprites, fn(sprite) ->
    JS.update(sprite, %{ "rotation" => sprite.rotation + 0.1 })
  end)
end

defp init() do
  Elixir.Agent.start(fn -> %{ "sprites" => [] } end, [name: __MODULE__])
end

defp sprites() do
  Elixir.Agent.get(__MODULE__, fn(x) -> x.sprites end)
end

defp add_sprite(sprite) do
  Elixir.Agent.update(__MODULE__, fn
    (x) -> %{ x | "sprites" => x.sprites ++ [sprite] }
  end)
end
```

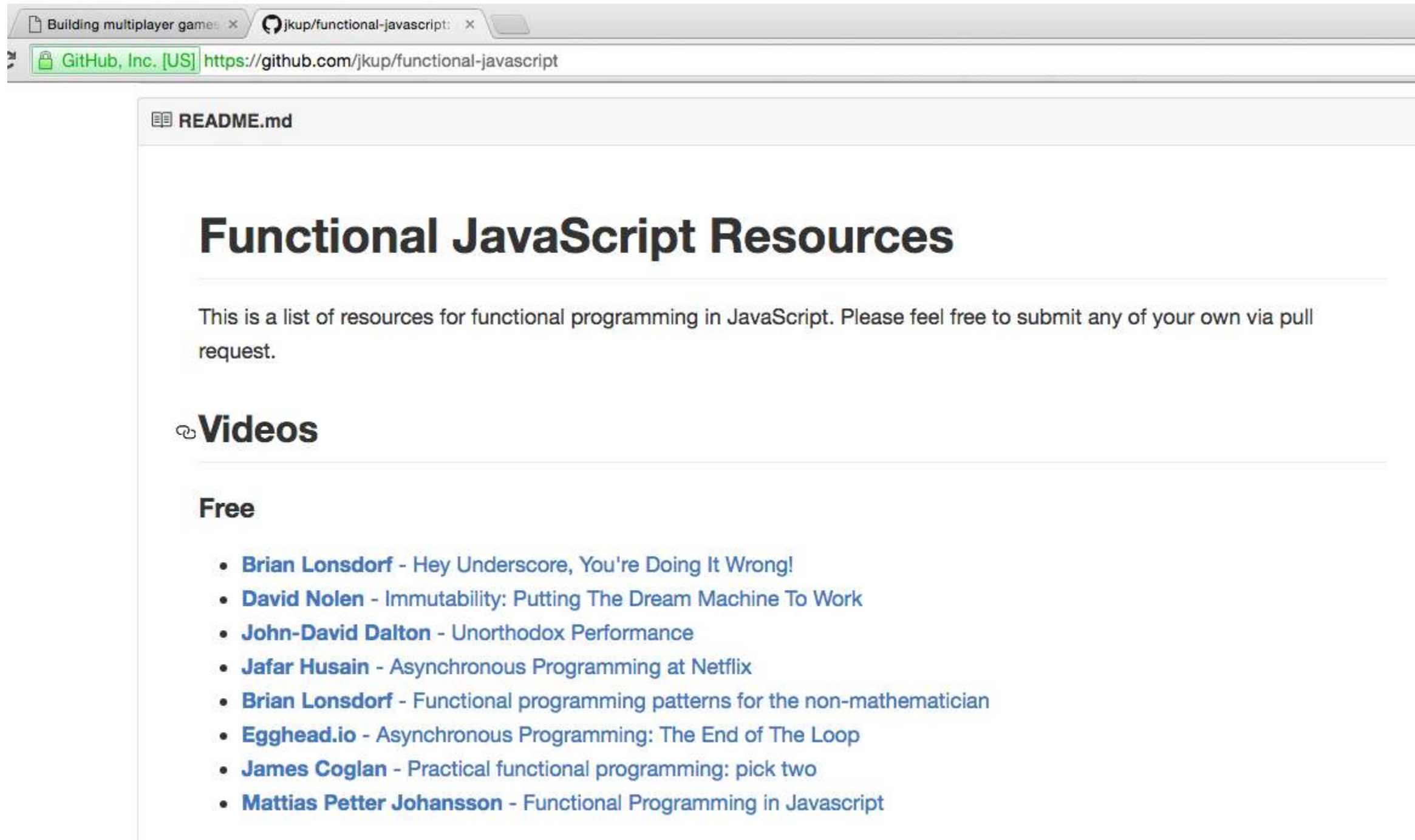


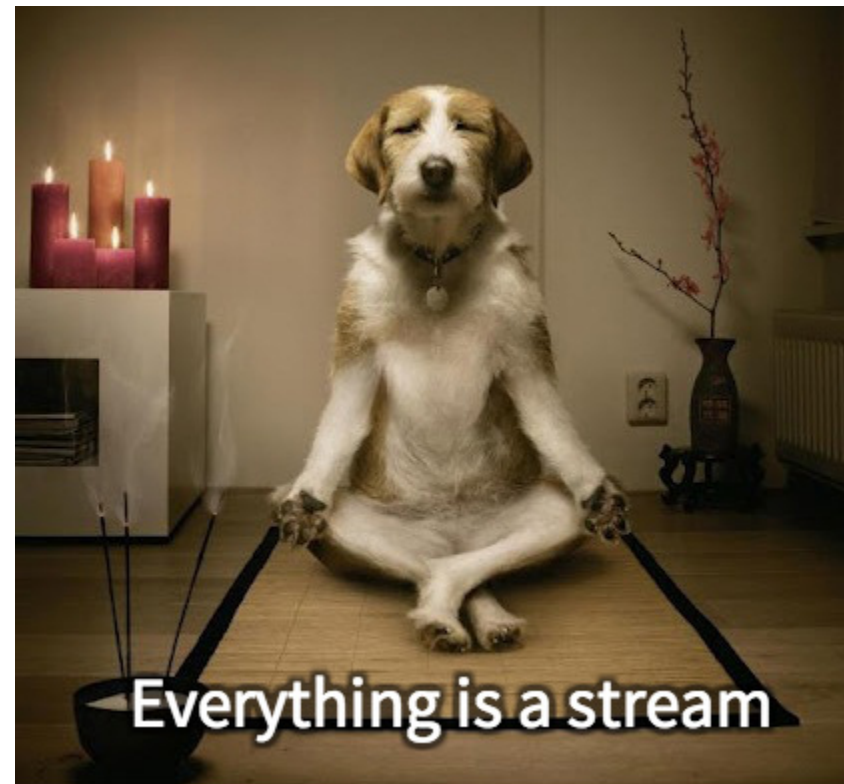
Phoenix Framework

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Hello
From
Elixirscript

Functional Javascript





ramda.js or kefir.js

Elm

Hex Package Generators

Phoenix Presence

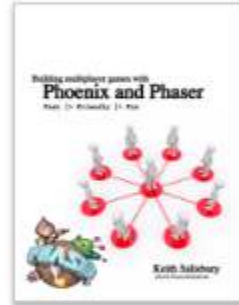
Explore Gamification

- Turn based games
- Winning/Losing
- Levels
- Challenges
- Virtual Goods
- Leaderboards
- Power/Energy
- High Scores
- Badges
- Authentication/Sign in
- Social sign in - Facebook/Twitter/Oauth
- Gestures on mobile (<http://hammerjs.github.io/>)

Phaser 3 aka Lazer



Store Read Write Support Blog



Building multiplayer games with Phoenix and Phaser

Overview

Getting Started

Unload Book Cover

Overview

Title: Building multiplayer games with Phoenix and Phaser

Authors: Keith Salisbury

Teaser Text: Phoenix and Phaser might sound like a cheesy pop band but get these two frameworks...

Categories: Functional Programming, Gaming, HTML5 and JavaScript

[Preview Book](#)

Thanks for Watching

Created by [Keith Salisbury](#) / [@ktec](#)